Umamahesh Manubolu

AI Research Engineer - ML Engineering

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3+ Years Experience in Machine Learning, Computer Vision & Robotics Research

M.Sc. Mechatronics | Python Expert & C++ | English C1 | Germany

Summary

Research-focused AI Engineer with 3+ years in Machine Learning, Computer Vision, Cognitive Robotics, and Intelligent Perception in Technical Systems. Strong mathematical foundations with advanced Python and C++ programming skills for research and development. Designed scalable machine learning and transformer models for multimodal perception and robotics systems. Highly Motivated to building something new.

Professional Experience

Master Thesis - Hybrid Machine Learning Systems

Universität Siegen | NRW, Germany — Mar 2025 – Oct 2025

- Designed and evaluated hybrid Markov Chain + Seq2Seq LSTM architecture from scratch using PyTorch, for predictive modelling technical systems, improving event prediction coverage by 6%.
- Conducted structured research experiments using rigorous mathematical modelling and probabilistic reasoning, strengthening foundations in applied mathematics and analytical problem-solving.
- Implemented scalable Python-based research pipelines for data preprocessing, tokenization, normalization, and evaluation on Time-Triggered dataset.
- Executed distributed training on SLURM-based HPC clusters, reducing model training time by 35%.
- Logged and analysed 50+ experimental runs using Weights & Biases, ensuring reproducibility for academic research publication standards.

Student Assistant – Deep Learning & Embedded Systems

Universität Siegen | NRW, Germany — Aug 2024 – Feb 2025

- Developed 12+ deep learning architectures using PyTorch (Python) for sequential prediction tasks in Intelligent Technical Systems.
- Improved model performance by 6% through hyperparameter optimization and systematic error analysis using strong analytical and problem-solving skills.
- Supported faculty-led research projects in machine learning and robotics, collaborating within a 3-member research team.
- Implemented Bayesian optimization workflows for automated hyperparameter search.
- Deployed ML pipelines on Linux-based HPC infrastructure using SSH and Docker, enhancing reproducibility by 30%.

AI Research Intern - Multimodal Perception & Tracking

Fraunhofer IVI | Ingolstadt, Germany — Nov 2023 – Apr 2024

- Implemented multimodal sensor fusion pipeline integrating LiDAR + camera data for multi-object tracking.
- Developed Kalman/UKF filtering algorithms and transformer-based decoder models evaluated on 1M+ KITTI-2D/3D Dataset.
- Containerized full ML training and inference pipelines using Docker for scalable cloud and edge deployment.
- Optimized GPU batch processing and memory utilization, reducing inference latency by 25%.

- Deployed containerized pipeline across multiple environments with 0 environment conflicts.

Projects

AI-Powered Resume and Cover Letter Tailoring Website

- Developed a web app with LangChain, RAG, Pinecone, LLMs, and Next.js, deployed on Vercel, reducing resume tailoring effort by 70%.
- Built and deployed an AI-powered resume system using cloud pipelines and prompt design, cutting tailoring time by 50% and improving application success by 30%.

RGB-D Object Detection - Computer Vision for Intelligent Perception Jun 2024 – Dec 2024

- Built end-to-end computer vision pipeline using YOLOv8 for RGB-D perception systems, achieving 84.3% mAP.
- Processed 5k+ warehouse images using structured annotation and augmentation pipelines in Python.
- Deployed optimized model on NVIDIA Jetson Nano (edge robotics system), achieving 100ms inference latency.
- Reduced inference time to 100–150ms per frame on edge GPU hardware.
- Increased dataset robustness by implementing ±10% lighting and occlusion variations.

Keywords: Computer Vision, Intelligent Perception, Robotics.

ConvLSTM Activity Recognition Pipeline Debugging

- Debugged unstable ConvLSTM training pipeline in Python.
- Fixed normalization, validation leakage, and batching inconsistencies.
- Stabilized training convergence and eliminated validation drift.
- Improved training reliability from inconsistent runs to 100% reproducibility.

4-DOF Robotic Manipulator - Motion Control System Oct 2021 – Mar 2022

- Designed and implemented inverse kinematics and motion planning algorithms in Python & MATLAB.
- Applied mathematical modelling and control theory for robotic arm precision improvement of 15%.
- Demonstrated expertise in Cognitive Robotics and Intelligent Technical Systems through system integration and testing.

Keywords: Cognitive Robotics, Robotics, Mathematical Foundations.

EDUCATION

M.Sc. Mechatronik - Universität Siegen, NRW, Germany Oct 2020 – Oct 2025

Focus: Machine Learning, Computer Vision (Deep Learning), Robotics, Intelligent Systems

B.Tech. Maschinenbau - CVR College of Engineering (JNTU-H), India Jun 2015 – May 2019

Technical Skills

- **Programming & AI Skills** Python (Advanced), C++, PyTorch, Linux, TensorFlow, Transformer Architectures, Seq2Seq, LSTM, CNN, Attention Mechanisms, Multimodal Learning, Sensor Fusion, Custom Loss Functions, Distributed Training Loops
- **Core Areas** Machine Learning, Computer Vision, Cognitive Robotics, Intelligent Perception, Generative AI, Transformer Models, model deployment.
- **Research Expertise** Mathematical Modelling, Analytical Problem-Solving, Experimental Design, Research Validation, ML frameworks.
- **Infrastructure** HPC Clusters, SLURM, Docker, AWS (Cloud), Render, Vercel (Frontend)

CERTIFICATIONS

Machine Learning Specialization (Coursera) | Deep Learning (Udemy) | IBM Generative AI & Transformer Models (Coursera)